MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 05** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 13: Statistics** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 05

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In step-deviation method, $u_i = \dfrac{x_i - a}{h}$, where h is assumed to be: [1]

(A) any real number (B) common factor of class marks
(B) common class size of the intervals (D) total frequency

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) common class size of the intervals
Justification: In the step-deviation method for grouped data having uniform class width, h denotes the common class size (width) of the intervals (h = Upper Limit − Lower Limit).

Q2. If the mean of first n natural numbers is $\dfrac{5n}{9}$, then the value of n is: [1]

(A) 5 (B) 7 (C) 9 (D) 10

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (C) 9

Justification: The sum of the first n natural numbers is $\sum_{i=1}^{n} i = \dfrac{n(n+1)}{2}$.

$$\text{Mean} = \frac{\sum i}{n} = \frac{n(n+1)}{2n} = \frac{n+1}{2}.$$

Given Mean $= \dfrac{5n}{9}$:

$$\frac{n+1}{2} = \frac{5n}{9} \implies 9(n+1) = 10n \implies 9n + 9 = 10n \implies \boxed{n = 9}.$$

Q3. For a distribution, if $\sum f_i = 100$, assumed mean $a = 25$, $\sum f_i d_i = 20$, then the mean is: [1]

(A) 25.2 (B) 24.8 (C) 27.0 (D) 25.0

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 25.2

Justification:

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i} = 25 + \frac{20}{100} = 25 + 0.2 = \mathbf{25.2}.$$

$$\boxed{\bar{x} = \mathbf{25.2}}$$

Q4. Which of the following measures of central tendency is least affected by extreme values (outliers)? [1]

(A) Arithmetic Mean (B) Median (C) Mode (D) Range

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Median

Justification: The median is a positional measure of central tendency based entirely on the middle rank of observations; hence, large outlying values at the extremes have no impact on its value.

Q5. **Assertion (A):** The median of a continuous grouped distribution can be computed without altering open-ended outermost classes, provided the median class itself is not an open-ended class.

Reason (R): In the formula Median $= l + \left(\frac{N/2 - cf}{f}\right) h$, the value depends strictly on the median class boundaries, half the total frequency $(N/2)$, and the preceding cumulative frequency (cf). [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Calculating the median requires only the knowledge of total items N, the preceding cumulative frequency cf, and the lower limit l, frequency f, and class width h of the median class. The open-ended extremities do not enter the formula. Thus, both statements are true and Reason (R) correctly explains Assertion (A).

SECTION B: Very Short Answer Type

MATHS MANTHAN | $(1 \times 2 = 2$ Marks)

Q6. If the median of a distribution is 26 and its mean is 24, find the mode using the empirical relationship. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Quoting the Empirical Relationship [1 Mark]

$$\text{Mode} = 3(\text{Median}) - 2(\text{Mean})$$

Step 2: Substitution & Calculation [1 Mark]

Given Median $= 26$ and Mean $= 24$:

$$\text{Mode} = 3(26) - 2(24) = 78 - 48 = 30.$$

$$\boxed{\text{Mode} = \mathbf{30}}$$

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. Find the mode of the ages of 80 patients admitted to a clinic: **[3]**

Age (years)	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
Patients	12	18	22	16	8	4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Modal Class & Parameter Identification [1 Mark]

The maximum frequency is 22, corresponding to the class **30 – 40**.

$\therefore$ Modal Class $= 30 - 40$.

Lower limit $(l) = 30$, class size $(h) = 10$.

$f_1 = 22$ (modal frequency), $f_0 = 18$ (preceding frequency), $f_2 = 16$ (succeeding frequency).

Step 2: Formula Declaration & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 30 + \left(\frac{22 - 18}{2(22) - 18 - 16}\right) \times 10 = 30 + \left(\frac{4}{44 - 34}\right) \times 10 = 30 + \left(\frac{4}{10}\right) \times 10$$

Step 3: Calculation & Final Result [1 Mark]

$$\text{Mode} = 30 + 4 = 34 \text{ years}$$

$$\boxed{\text{Mode of the ages} = \mathbf{34} \text{ years}}$$

Q8. If the mean of the following frequency distribution is 54, find the value of p: **[3]**

Class Interval	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
Frequency	7	p	10	9	13

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Direct Method Calculation Table [1.5 Marks]

Class Interval	Frequency (f_i)	Class Mark (x_i)	f_ix_i
$0-20$	7	10	70
$20-40$	p	30	$30p$
$40-60$	10	50	500
$60-80$	9	70	630
$80-100$	13	90	1170
Total	$\sum f_i = 39+p$		$\sum f_ix_i = 2370+30p$

Step 2: Equation Formulation with Given Mean [1 Mark]

$$\bar{x} = \frac{\sum f_ix_i}{\sum f_i} \implies 54 = \frac{2370+30p}{39+p}$$

$$54(39+p) = 2370+30p \implies 2106+54p = 2370+30p$$

Step 3: Solving for p & Final Boxed Answer [0.5 Mark]

$$54p-30p = 2370-2106 \implies 24p = 264 \implies p = \frac{264}{24} = 11.$$

Value of p = **11**

Q9. Find the median marks from the following cumulative distribution of 80 students: [3]

Marks	<10	<20	<30	<40	<50	<60
Students (cf)	7	17	37	57	75	80

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Step 1: Frequency Distribution & Median Class [1 Mark]

Class Interval	Frequency (f_i)	Cumulative Frequency (cf)
$0-10$	7	7
$10-20$	$17-7=10$	17
$20-30$	$37-17=20$	37
$30-40$	$57-37=20$	57
$40-50$	$75-57=18$	75
$50-60$	$80-75=5$	80
Total	$N=\sum f_i = 80$	

Step 2: Median Class Parameters & Formula [1 Mark]

$N = 80 \implies \frac{N}{2} = 40.$

Cumulative frequency just greater than 40 is 57, corresponding to class **30 − 40**.

$\therefore$ Median Class $= 30-40$.

Lower limit $(l) = 30$, cf of preceding class $= 37$, frequency $(f) = 20$, class size $(h) = 10$.

Step 3: Calculation & Final Result [1 Mark]

$$\text{Median} = l + \left(\frac{\frac{N}{2}-cf}{f}\right)\times h = 30 + \left(\frac{40-37}{20}\right)\times 10 = 30+\frac{30}{20} = 30+1.5 = 31.5$$

Median Marks = **31.5**

SECTION D: Case Study Based Question

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. CASE STUDY: ANNUAL INTER-SCHOOL ATHLETICS 100M SPRINT TIMING ANALYSIS [4 Marks]

During an inter-district track and field meet, the race completion timings of 40 sprinters in the 100m heats were recorded using automated digital laser timing gates.

Time (seconds)	$13.5 - 14.0$	$14.0 - 14.5$	$14.5 - 15.0$	$15.0 - 15.5$	$15.5 - 16.0$
Number of Sprinters	4	8	15	11	2

(i) Find the class mark and width of the modal sprint timing class. [1]

(ii) Calculate the median race completion time of the athletes. [2]

(iii) How many sprinters finished the 100m dash in under 15.0 seconds? [1]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Modal Class Mark & Width [1 Mark]

The highest frequency is 15, which corresponds to the class $\mathbf{14.5 - 15.0}$.
$\therefore$ Modal Class $= 14.5 - 15.0$ seconds.

$$\text{Class Mark } (x_i) = \frac{14.5 + 15.0}{2} = \frac{29.5}{2} = \mathbf{14.75} \text{ seconds.}$$

$$\text{Class Width } (h) = 15.0 - 14.5 = \mathbf{0.5} \text{ seconds.}$$

$$\boxed{\text{Class Mark} = \mathbf{14.75} \text{ s}, \quad \text{Class Width} = \mathbf{0.5} \text{ s}}$$

Subpart (ii): Median Race Completion Time [2 Marks]

Total sprinters $N = 40 \implies \frac{N}{2} = 20$.
Cumulative frequencies: $13.5 - 14.0 : 4$; $14.0 - 14.5 : 12$; $14.5 - 15.0 : 27$; $15.0 - 15.5 : 38$; $15.5 - 16.0 : 40$.
The cumulative frequency just greater than 20 is 27, corresponding to median class $\mathbf{14.5 - 15.0}$.
Here, $l = 14.5$, $cf = 12$, $f = 15$, $h = 0.5$.

$$\text{Median} = l + \left(\frac{\frac{N}{2} - cf}{f} \right) \times h \text{[1 Mark]}$$

$$\text{Median} = 14.5 + \left(\frac{20 - 12}{15} \right) \times 0.5 = 14.5 + \frac{4}{15} \approx 14.5 + 0.267 = \mathbf{14.77} \text{ s} \quad \left(\text{or } 14\frac{23}{30} \text{ s} \right) \text{[1 Mark]}$$

$$\boxed{\text{Median Time} = \mathbf{14.77} \text{ seconds} \quad \left(\text{or } \mathbf{14\frac{23}{30}} \text{ seconds} \right)}$$

Subpart (iii): Sprinters Finishing in Under 15.0 Seconds [1 Mark]

Sprinters with timing < 15.0 seconds comprise intervals $[13.5 - 14.0]$, $[14.0 - 14.5]$, and $[14.5 - 15.0]$:

$$\text{Total Sprinters} = 4 + 8 + 15 = \mathbf{27}.$$

$$\boxed{\text{Number of sprinters with time } < 15.0 \text{ s} = \mathbf{27}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★